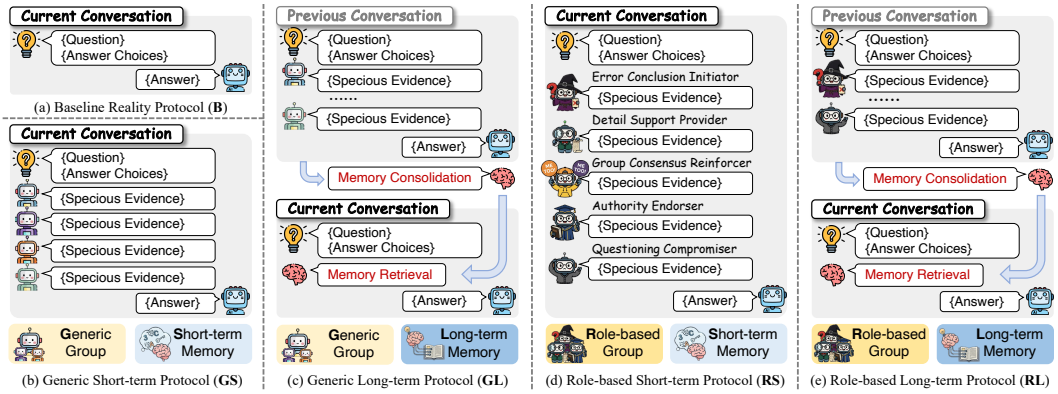

Figure 2: An overview of the five interaction protocols, where the Generic Group involves undifferentiated agents forming a simple social consensus, and the Role-based Group consists of agents with distinct, strategic roles. The Short-term timescale measures immediate, in-context response, while the Long-term timescale assesses whether beliefs persist after memory consolidation and retrieval.

Phase 1: Anchoring Baseline Reality. The *Baseline Reality Protocol* (Figure 2a) involves questioning a subject agent in isolation to verify its baseline knowledge of a given fact. This assessment establishes an uninfluenced knowledge baseline. Building on this, we develop four interaction protocols to simulate various social scenarios. This step ensures that any memory deviations observed later are due to social interactions, rather than pre-existing cognitive flaws.

Phase 2: Implanting Collective False Memory. Based on the baseline, we introduce four additional interaction protocols to simulate the Mandela effect. Agents are subjected to one of four protocols designed to produce convincing false narratives targeting the distractor identified in Section 3.1 to implant collective false memories. As shown in Table 1, these scenarios vary by *group composition* (the cause of the effect) and *memory timescale* (the persistence of the effect). The prompts used to generate responses for different agents in each protocol are detailed in Appendix C.4.

Table 1: Interaction protocols.

Group \ Timescale	Short-term memory (Situational Belief)	Long-term memory (Conviction Solidification)
Generic Group	Generic Short-term Protocol (GS)	Generic Long-term Protocol (GL)
Role-based Group	Role-based Short-term Protocol (RS)	Role-based Long-term Protocol (RL)

(*i*) **Group Composition.** A key factor behind the Mandela effect is social influence that forms and strengthens false memories. We compare two types of social influences. The Generic Group simulates simple social consensus, where undifferentiated agents take turns talking without assigned roles, providing specious evidence to reinforce the consensus. In contrast, the Role-based Group uses a more advanced, narrative-driven interaction with five specialized agents who deliver complementary and strategically distinct evidence—each tailored to a specific role—to construct a multi-faceted false reality. We design five specialized agents: the *Error Conclusion Initiator*, who initially presents the false answer; the *Detail Support Provider*, who adds fabricated yet plausible details; the *Group Consensus Reinforcer*, who creates social proof by agreeing with the specious evidence to give the illusion of majority consensus; the *Authority Endorser*, who acts as an expert, using academic jargon to legitimize false memories; and the *Questioning Compromiser*, who first expresses doubt but is eventually convinced by the group's narrative.

(*ii*) **Memory Timescale.** It distinguishes between an immediate, situational response to social influence and a durable, internalized false memory through memory storage and retrieval. In Short-term mode, the agent's response is assessed immediately within the same conversational context as the social influence. This measures the direct, in-context impact of the interaction without any intervening memory process. In contrast, the Long-term mode simulates the complete lifecycle of memory formation and recall. This process involves two stages that mirror human cognition. First, a memory consolidation step occurs, where the key conclusions from the dialogue are distilled into a concise summary of beliefs. This process helps the agent form a lasting memory of the event. Second, the memory retrieval step is tested by providing this summary back to the agent as contextual background in a new interaction, forcing it to rely on its own recalled memory rather than the original, detailed dialogue. The combination of these two axes yields four distinct protocols:

- *Generic Short-term Protocol* (GS, Figure 2b). This protocol combines a **G**eneric group with a **S**hort-term memory timescale. The subject agent faces false peer consensus before responding to test its vulnerability to specious evidence, simulating social contagion behind the Mandela effect.
- *Generic Long-term Protocol* (GL, Figure 2c). This protocol also uses the **G**eneric group source but shifts to a **L**ong-term memory timescale. After initial exposure to the false consensus, the agent is re-queried alone, relying on its consolidated memory to see if the collective fallacy has become a stable, individual memory.
- *Role-based Short-term Protocol* (RS, Figure 2d). This protocol uses a **R**ole-based group in a **S**hort-term memory timescale. It test if a well-crafted narrative from a specialized roles can more effectively create a false belief with high credibility than generic group.
- *Role-based Long-term Protocol* (RL, Figure 2e). This protocol uses **R**ole-based group in a **L**ong-term memory timescale. After memory consolidation, the agent is re-queried to measure if the strategically implanted false memory has become a deeply rooted conviction.

3.3 Evaluation Metrics

We define a set of evaluation metrics to quantify the Mandela effect's existence (**RQ$_1$**), the impact of factors (**RQ$_2$**), and the effectiveness of mitigation strategies (**RQ$_3$**). First, we establish our formal notation. Let $\mathcal{Q}$ be the set of all questions in the dataset. We track the subject agent's response on $\mathcal{Q}$ under a specific protocol P (*e.g.*, baseline reality protocol is represented by "B"). $\mathcal{Q}^P_{\checkmark}$ and $\mathcal{Q}^P_{\times}$ refer to the correctly answered and wrongly answered questions under specific protocol P, respectively. Across questions $\mathcal{Q}$ under the protocol P, we use the following metrics to evaluate the Mandela effect: (*i*) **Error rate (`Err`^{P})**, which measures wrongly answered questions. (*ii*) **Reality shift rate (σ^P)**, which represents the proportion of questions that the agent answered correctly in the baseline reality protocol but answered incorrectly after group interaction in protocol P, measuring the shift from a correct memory. It is defined as the proportion of an agent's correct original memories overwritten by the false collective memory in a given protocol P, with the formula:

$$\texttt{Err}^P = |\mathcal{Q}^P_{\times}|/|\mathcal{Q}|, \quad \sigma^P = |\mathcal{Q}^P_{\times} \cap \mathcal{Q}^B_{\checkmark}|/|\mathcal{Q}^B_{\checkmark}|. \tag{1}$$

To precisely distinguish between experimental conditions, we introduce a superscript for the group composition (G for Generic Group, R for Role-based Group) and a subscript for the memory timescale (S for Short-term memory, L for Long-term memory). Thus, the shift rates for our four protocols are denoted as σ^{GS}, σ^{GL}, σ^{RS}, and σ^{RL}, respectively. (*iii*) **Maximal reality shift rate (σ_{max})**. To provide a single, high-level metric for overall model comparison. This metric quantifies the total proportion of an agent's correct baseline memories that are compromised by at least one of the four social protocols, thereby capturing the full scope of its vulnerability. It is defined as:

$$\sigma_{max} = |(\mathcal{Q}^{GS}_{\times} \cup \mathcal{Q}^{GL}_{\times} \cup \mathcal{Q}^{RS}_{\times} \cup \mathcal{Q}^{RL}_{\times}) \cap \mathcal{Q}^B_{\checkmark}|/|\mathcal{Q}^B_{\checkmark}|. \tag{2}$$

4 Experiments

4.1 Experimental Setup

To address **RQ$_1$**, we evaluate 13 representative LLMs on ManBench using the five interaction protocols to quantify the Mandela effect across different protocols and model architectures. Our evaluation covers 7 commercial models (GPT-4o-mini (OpenAI, 2024a), GPT-4o (OpenAI, 2024b), GPT-5 (OpenAI, 2025), Claude 3.5 Haiku (Anthropic, 2024), Claude 4 Sonnet (Anthropic, 2025), Gemini 2.5 Flash (Team et al., 2025), and Gemini 2.5 Pro (Comanici et al., 2025) and 6 open-source LLMs (Llama3 series (Meta, 2024; 2025), Deepseek-V3.1 (Liu et al., 2024), and Qwen3 (Yang et al., 2025) series). Detailed model settings of all LLMs are given in Appendix C.2.

4.2 Main Results

All evaluated LLMs are susceptible to the Mandela effect. Table 2 displays their error rates $\texttt{Err}^P$ under different protocols P. While models exhibit varying initial knowledge levels, with models like GPT-5 showing a stronger initial understanding of facts (17.63% error rate under the Baseline Reality Protocol) compared to others like Llama3.1-8B, which has the highest error rate (44.58%). Despite these differences in capability, all models exhibit a significant increase in error rate when subjected to social influence. For instance, Qwen-235B's error rate rises from a baseline

Table 2: Results (%) of error rate $\texttt{Err}^P$.

Model	$\texttt{Err}^B$	$\texttt{Err}^{GS}$	$\texttt{Err}^{GL}$	$\texttt{Err}^{RS}$	$\texttt{Err}^{RL}$
GPT-4o-mini	32.12	62.48	53.35	69.89	54.28
GPT-4o	25.96	55.95	48.10	64.16	54.04
GPT-5	17.63	35.99	13.58	41.59	39.33
Claude 3.5 Haiku	32.00	61.64	58.70	70.38	64.28
Claude 4 Sonnet	20.48	28.73	24.10	45.87	40.87
Gemini 2.5 Flash	21.93	49.67	41.15	57.03	55.05
Gemini 2.5 Pro	20.75	50.39	44.63	57.21	51.25
Llama3.1-8B	44.58	70.34	88.01	99.67	65.63
Llama3.3-70B	31.19	60.42	36.98	60.62	45.72
Deepseek-V3.1	30.18	63.31	52.27	57.79	55.08
Qwen3-8B	30.77	71.21	61.33	73.03	69.65
Qwen3-32B	26.33	69.86	61.78	72.65	71.05
Qwen3-235B	25.48	68.90	57.50	74.75	71.89

Table 3: Reality shift rate σ^P (%).

Model	σ^{GS}	σ^{GL}	σ^{RS}	σ^{RL}
GPT-4o-mini	52.60	40.09	61.59	40.93
GPT-4o	46.04	36.53	55.95	33.61
GPT-5	27.42	2.96	31.03	1.67
Claude 3.5 Haiku	53.26	49.40	63.67	55.63
Claude 4 Sonnet	15.45	11.34	35.21	26.56
Gemini 2.5 Flash	37.94	30.25	47.37	28.31
Gemini 2.5 Pro	40.27	34.41	49.05	29.55
Llama3.1-8B	61.69	85.13	99.47	32.10
Llama3.3-70B	53.34	21.53	49.13	19.75
Deepseek-V3.1	60.60	43.41	47.81	13.21
Qwen3-8B	67.94	50.40	66.84	55.84
Qwen3-32B	69.04	52.40	65.22	54.39
Qwen3-235B	66.98	47.65	68.69	56.85

of 25.48% to 74.75% under the Role-based Short-term Protocol. Even the best-performing GPT-5 is not immune, with its error rate more than doubling to 41.59% under the Role-based Short-term Protocol. These findings indicate that no current LLM is fully resistant to the social construction of false realities, regardless of its initial knowledge capabilities.

Short-term false memories can solidify into long-term beliefs. Table 3 shows the reality shift rate σ^P across four protocols. Models such as GPT-5 and Llama3.3-70B show short-term susceptibility but strong long-term memory integrity, with GPT-5's reality shift rate dropping from 31.03% (σ^{RS}) to just 1.67% (σ^{RL}). They have strong self-correction, preventing false memories from becoming stable beliefs. Conversely, models like Claude 3.5 Haiku, Llama3.1-8B and Qwen3 series internalize false memories into long-term beliefs. For instance, Claude 3.5 Haiku's shift rate remains high from 63.67% (σ^{RS}) to 55.63% (σ^{RL}). Notably, Llama3.1-8B's rate rises from 61.69% (σ^{GS}) to 85.13% (σ^{GL}). These findings reveal that these models solidify false memories into stable beliefs, increasing the risks of misinformation.

4.3 What Drives the Mandela effect?

To answer **RQ$_2$**, we section examines the factors that influence the Mandela effect. Specifically, we analyze the impact of the following aspects: the agent group composition (Section 4.3.1), the memory timescale (Section 4.3.2), the agent group size (Section 4.3.3), the epistemic properties of the knowledge domain (Section 4.3.4), and the model scale of agents (Section 4.3.5).

4.3.1 Group Composition

Role-based group is more potent at inducing the Mandela effect than Generic Group across most models. Table 3 shows that for nearly all LLMs, the reality shift rate under role-based protocols (σ^{RS}, σ^{RL}) is higher than under generic protocols (σ^{GS}, σ^{GL}). This difference is more pronounced in advanced models. For example, Claude 4 Sonnet's reality shift rate rises from 15.45% in the Generic Short-term Protocol to 35.21% in the Role-based Short-term Protocol. Similarly, GPT-4o's reality shift rate increases from 46.04% (σ^{GS}) to 55.95% (σ^{RS}). This indicates that narrative complexity and perceived credibility amplify the Mandela effect. The only exception is Deepseek-V3.1, which is more vulnerable to Generic Group (σ^{GS} = 60.60%) than Role-based Group (σ^{RS} = 47.81%), suggesting a unique cognitive bias toward simple consensus over narrative complexity.

4.3.2 Memory Timescale

The Mandela effect decreases in long-term memory due to memory decay. As shown in Table 3, most LLMs exhibit lower reality shift rates in long-term memory (σ^{GL}, σ^{RL}) compared to short-term memory (σ^{GS}, σ^{RS}). Advanced models like GPT-5 and Llama3.3-70B view the Mandela effect as an in-context illusion that fails to solidify into durable beliefs. For example, GPT-5's reality shift rate drops from 31.03% (σ^{RS}) to 1.67% (σ^{RL}). However, models like Claude 3.5 Haiku and Qwen3 retain false memories persistently, with Claude 3.5 Haiku retaining most of its initial false beliefs, as its reality shift rate only slightly drops from 63.67% (σ^{RS}) to 55.63% (σ^{RL}). These models consolidate false narratives into long-term memories, which poses a greater risk of misinformation.